

Operations Research
Problem Set 1

Problem 1

A small pottery studio makes two items for the weekly market: a decorated mug and a plain bowl. Each mug uses 1 kg of clay and 3 dl of glaze (it is dipped twice and hand-painted); each bowl uses 2 kg of clay but only 1 dl of glaze. This week the studio has 80 kg of clay and 90 dl of glaze, and nothing else is in short supply. The net margin is 30 DH per mug and 40 DH per bowl, and everything produced is sold.

- (a) Define the decision variables and write the linear program that maximises the weekly margin.
- (b) Put the program into equational form. How many variables n and equations m does it have? Verify that A has full row rank.
- (c) There are $\binom{4}{2} = 6$ candidate bases. For each one, compute the basic solution and state whether it is feasible.
- (d) Sketch the feasible region in the plane and match each basic feasible solution with a vertex. Which one is optimal?

Problem 2

A dairy farm mixes a daily ration for each cow from two feeds: barley meal at 5 DH/kg and alfalfa pellets at 8 DH/kg. One kilogram of barley meal supplies 2 units of digestible protein and 1 unit of fibre; one kilogram of alfalfa pellets supplies 1 unit of protein and 3 units of fibre. The vet requires at least 8 units of protein and at least 9 units of fibre per cow per day. The farmer wants the cheapest ration.

- (a) Write the linear program.
- (b) Convert it to equational form, as a maximisation. Explain why a slack variable is not what you add here, and what the new variables mean in the language of the farm.
- (c) Compute the basic solution belonging to the basis of the two surplus variables. What do you observe, and why does this matter for a method that must start from a basic feasible solution?
- (d) Enumerate all six bases, identify the basic feasible solutions, and find the cheapest ration.

Problem 3

A 100-hectare irrigated farm must decide what to do with every hectare this season: plant wheat, plant sunflower, or leave the land fallow under a set-aside scheme. Wheat consumes 3 units of water per hectare and sunflower 2; fallow land consumes none. The seasonal water allocation is 240 units. A supply contract already signed obliges the farm to plant at least 20 ha of wheat. Net margins are 4, 5 and 1 thousand DH per hectare for wheat, sunflower and fallow respectively.

- (a) Write the linear program. Note carefully which constraint is an equality, and why.
- (b) Remove the lower bound $x_1 \geq 20$ by a change of variable, then put the program into equational form.

- (c) Enumerate all bases of the equational form and list the basic feasible solutions.
- (d) Give the optimal land allocation in the original variables and the resulting margin. Which resources are binding?

Problem 4

A print shop has to deliver 6000 posters by the end of the day and owns three presses. Press A produces 500 posters per hour, press B 300, and press C 200. The presses run in parallel, so the job is finished when the slowest of the three finishes its share. Press C has only one ink cartridge left, good for at most 1000 posters. The manager wants to finish as early as possible.

- (a) Explain why “minimise the largest of three quantities” is not linear as stated, and show how one extra variable t turns the model into a linear program. Write the linear program.
- (b) Put the linear program into equational form and state n and m .
- (c) Consider the plan $x = (3125, 1875, 1000)$ with $t = 6.25$. Check that it is feasible, exhibit the m columns that form a basis for it, prove that they are linearly independent, and say whether the basic feasible solution is degenerate.
- (d) What would the optimal completion time be if press C had unlimited ink? Interpret the difference.

Problem 5

A workshop assembles solar water heaters. Firm orders are 100 units for March, 180 for April and 120 for May. The assembly line can produce at most 150 units in any month. Production costs 3000 DH per unit in every month. A unit finished early and held in the warehouse costs 200 DH per month; a unit delivered late costs the workshop a contractual penalty of 500 DH per month of delay. The warehouse is empty at the start of March, and all orders must be settled by the end of May.

- (a) Model the situation with production variables $p_t \geq 0$ and inventory variables I_t that are unrestricted in sign, a negative I_t meaning a backlog. Write the balance equations and the objective.
- (b) Eliminate the free variables and put the model into equational form. How many variables and equations result?
- (c) Verify that the plan $p = (130, 150, 120)$ with $I = (30, 0, 0)$ is a basic feasible solution: identify the basic variables, check that there are m of them and that the corresponding columns are independent, and say whether it is degenerate. Compute its cost.
- (d) The penalty for late delivery is renegotiated down to 100 DH per unit per month. Re-solve by inspection and state the new plan and cost.

Problem 6

This problem collects the theoretical threads of the preceding five.

- (a) For the ceramics studio of Problem 1, verify directly that the map sending a basic feasible solution to (x_1, x_2) is a bijection onto the vertices of the feasible polygon. Which geometric fact corresponds to a variable being non-basic?
- (b) Problems 1, 2 and 3 each had $\binom{4}{2} = 6$ bases but 4, 3 and 4 basic feasible solutions respectively. Explain the distinct reasons for which a basis may fail to contribute a new basic feasible solution.
- (c) Let $P = \{x : Ax = b, x \geq 0\}$ be non-empty, with A of full row rank. Show that P contains at least one basic feasible solution. *Hint:* take a feasible x with the fewest positive entries, and show that if the columns indexed by its support were dependent, you could move along a null-space direction and zero out one more entry.